Movement Logger — Race Mode Setup

Live rider tracking: every rider's phone streams its GPS position to one shared map on the desktop.

Requirements

- All phones and the desktop computer on the same WiFi network (club/venue WiFi works fine).
- Versions: Desktop app 0.0.68 or newer, Android app 0.0.53 or newer, iOS app 1.0.31 or newer.
- No extra hardware required: phones use their own GPS; Android can also use a u-blox USB receiver.

1 — Desktop (race committee)

1. Install: github.com/zdavatz/movement_logger_desktop/releases (macOS DMG, Windows zip, Linux tar).
2. Open the app -> Race tab -> click "Start listening".
3. The header now shows: Phones send to <IP>:47777 — riders enter exactly these two values.
4. Map controls: scroll or double-click or the -/+ buttons to zoom; "Satellite" switches to imagery; "Follow riders" keeps everyone in view automatically. macOS firewall prompt: click Allow.

Every received fix is also logged to ~/.movementlogger/race/race_<timestamp>.csv for post-race analysis.

2 — Android rider (u-blox USB receiver or phone GPS)

1. Install "Movement Logger" from Google Play: play.google.com/store/apps/details?id=ch.ywesee.movementlogger
2. GPS tab -> "Race mode" card: enter your rider name + the desktop IP and port from step 1.3.
3. Pick the GPS source: "u-blox USB" (plug the receiver into USB-C and tap Connect first) or "Phone GPS" (no extra hardware; allow the location permission) -> toggle ON.
4. "Sending — N fixes" counts up as soon as there is a GPS fix (needs sky view — go outside).

3 — iPhone / Apple Watch rider

1. Install "MovementLogger" from the App Store: apps.apple.com/app/id6769086271
2. GPS tab -> "Race mode" card: rider name + the desktop IP and port, then pick the GPS source:
   - iPhone GPS: toggle ON — done. The phone's own GPS starts automatically.
   - Apple Watch: toggle ON once with the iPhone nearby (hands the settings to the watch), then start a recording in the watch app. With the iPhone around, fixes relay through it; without it, the watch sends directly over the venue WiFi — watch-only riders work.

4 — Over the internet: the race relay (cellular race days)

WiFi only reaches ~50 m. For a real course, riders use mobile data and the organizer runs the relay (in every desktop release: race-relay-...-linux-musl.tar.gz) on any Linux server:

cp race-relay /usr/local/bin/ · cp race-relay.service /etc/systemd/system/
systemctl enable --now race-relay · open UDP 47777 · point a DNS name at the server.

Riders then enter the relay's host + 47777 instead of the desktop IP; the desktop ticks "Via relay". Set the same "Race token" on the desktop and every rider so strangers can neither watch nor appear on your map.

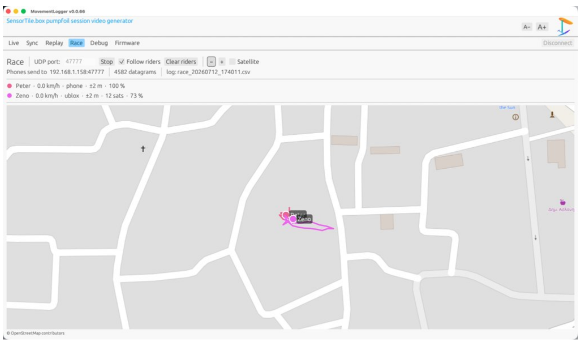
Reading the map

- Dot = the rider's latest position; the translucent circle around it is the receiver's own accuracy estimate (bigger circle = less precise fix). The coloured trail is the recently travelled path.
- The rider list shows: name, speed, source (ublox / phone / watch), +-accuracy, satellites, battery.
- A rider greyed out has sent nothing for 5 s — the last position stays on the map on purpose.

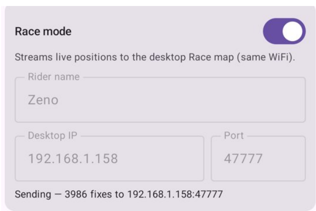
Troubleshooting

- No dot appears: same WiFi? IP and port typed exactly as the desktop shows? GPS fix yet (sky view)?
- Imprecise near buildings? Watch the accuracy circle — GPS is a 2-5 m instrument; open water is best.
- Two riders must not use the same rider name on the same source type.

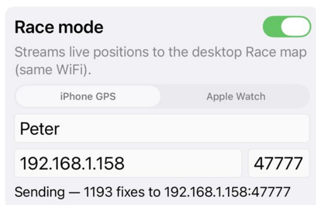
What it looks like



Desktop — Race tab



Android — Race mode card



iPhone — Race mode card

Desktop · **Android** · **iPhone**